

Olga Kondratenko

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Professional Summary

Machine Learning Engineer with 4+ years of experience building production AI systems and healthcare software in a medical-device and robotics R&D environment. Strong experience in machine learning, deep learning, computer vision, 3D medical imaging, and clinical data processing, with hands-on experience using PyTorch, TensorFlow, scikit-learn, MONAI, CUDA, VTK, and SimpleITK. Experienced developing ML models and data-processing pipelines for volumetric DICOM imaging, anatomical segmentation, image and metadata quality assessment, and quantitative image analysis. Strong software-engineering background spanning Python, C++, REST services, Docker, and Kubernetes, with experience taking ML research and prototypes into reliable production systems. Contributed to FDA submission studies, including training-data preparation and curation, and helped improve bone-segmentation accuracy. Member of the team recognized with the 2025 CEO Excellence Award for developing a pre-operative planning pipeline as part of a surgery-assisted robotic system.

Technical Skills

Programming Languages: Python, JavaScript, SQL, C++, Java

ML & AI: PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, LSTMs, Computer Vision, LLMs, Model Training, Model Evaluation, Inference Pipelines

GPU & Infrastructure: CUDA, GPU-Accelerated Computing, Docker, Kubernetes, Model Serving, Batching, Performance Profiling

Data & Scientific Computing: NumPy, pandas, SciPy, Apache Airflow, Kafka, Feature Engineering, Statistical Analysis

Software Engineering: FastAPI, Flask, REST APIs, Pydantic, OpenAPI/Swagger, Git, GitHub, Unit Testing, Integration Testing, Production Deployment

Medical AI: MONAI, VTK, SimpleITK, DICOM, 3D, Medical Imaging, Image Processing

Operating Systems: Linux, Ubuntu, macOS, WSL2, Windows

Professional Experience

Software Engineer – Robotics R&D

Smith+Nephew | December 2022 – July 2026

- Built and deployed production ML and AI systems in Python, translating research requirements and ambiguous engineering problems into reliable software for medical-device and robotics applications.
- Developed, trained, and evaluated ML and deep-learning models using PyTorch, TensorFlow, and scikit-learn, including 3D computer vision and medical-imaging models.
- Developed and deployed ML inference services using FastAPI and Flask, with hands-on experience with model serving, batching, and inference performance considerations.
- Used CUDA and GPU-accelerated computing for computationally intensive ML and image-processing workflows, with exposure to profiling and performance optimization.
- Containerized and deployed ML services using Docker and Kubernetes, including experience working with Kubernetes resource allocation and deployment constraints.

- Built scalable data-processing pipelines for large multimodal datasets and developed automated validation systems to detect malformed, incomplete, and low-quality inputs before downstream ML processing.
- Developed two ML models from scratch to assess whether incoming image pixel data and metadata were suitable for downstream ML pipelines.
- Worked with MONAI, VTK, SimpleITK, and DICOM for 3D medical-image processing, segmentation, and computer-vision workflows.
- Diagnosed issues across application code, ML pipelines, data processing, APIs, and deployment environments, collaborating with engineering and research teams to deliver production solutions.
- Expanded automated unit, integration, and functional testing, increasing unit-test coverage from 20% to 70%.
- Collaborated across engineering, research, and clinical teams to transition prototypes into validated production systems within a regulated medical-device environment.

Machine Learning Engineer (Volunteer)

MLDL Solutions | May 2022 – December 2022

- Performed exploratory data analysis and feature engineering on large environmental and geospatial datasets using Python, pandas, NumPy, and scikit-learn.
- Developed, trained, and evaluated machine-learning and deep-learning models using TensorFlow and PyTorch.
- Applied data preprocessing, feature engineering, model evaluation, and statistical analysis to large-scale datasets.
- Processed large-scale datasets using Google Earth Engine and Azure cloud services.
- Built interactive data-visualization dashboards using D3.js to communicate analytical findings and insights.

AI Researcher (Volunteer)

University of San Francisco – MAGICs Lab | September 2021 – December 2023

- Conducted machine-learning research involving data preprocessing, feature engineering, model development, and performance evaluation.
- Developed and evaluated deep-learning models for predictive analysis using Python, PyTorch, TensorFlow, and scientific-computing libraries.
- Processed and analyzed large datasets and developed analytical workflows supporting AI research.
- Created visualization and analysis tools to communicate experimental results and research findings.

Systems Administration Assistant

University of San Francisco | January 2022 – May 2022

- Installed, configured, and maintained Linux, Windows, and macOS systems supporting university teaching and research laboratories.
- Diagnosed and resolved hardware and software issues, minimizing downtime across university computing environments.

Education

University of San Francisco

Master of Science (M.S.), Computer Science | August 2020 – May 2022

Portfolio: <https://olga-kondr.github.io/>